# 1 Introduction

Large language models (LLMs) can translate natural-language goals into high-level robot actions [1, 2]. Long-running mobile robots must advance tasks while maintaining safety. Records of past actions, outcomes, and risks can complement current observations by informing risk assessment and updating empirical safety boundaries. Using this evidence over long deployments requires compact, traceable risk memory.

Experiential-learning methods reuse interaction histories through verbal reflection, natural-language experience, or reusable skills [3, 4, 5], whereas safe-planning methods bound robot decisions through temporal constraints, safety feedback, and formal guardrails [2, 6, 7]. Combining these directions for long-term risk-aware decisions introduces three requirements: compress growing histories at multiple abstraction levels, preserve evidence chains from abstract knowledge to source cases, and retrieve evidence aligned with each deliberative role. These requirements motivate our work.

We present a risk-experience-aware multi-role agent framework. Hierarchical risk memory compresses long-term experience, preserves evidence chains, and updates empirical safety boundaries. Role-conditioned retrieval supplies responsibility-relevant history to a Task Advocate, a Risk Critic, and a Safety Decision Agent. These agents produce only structured semantic recommendations; a deterministic constrained optimizer and an independent Safety Guard retain execution authority.

The main contributions of this work are:

1. We propose a risk-experience-aware multi-role agent architecture that provides responsibility-relevant history to three complementary roles while separating language reasoning from execution authority.
2. We develop hierarchical risk memory based on Risk Memory Cards. Its L0–L1–L2 representation captures environment, task, action outcome, failure cause, and risk factors while supporting compression, traceability, and safety-boundary evolution.
3. We design an evaluation protocol comprising multi-scenario closed-loop simulation, component ablations, and real-robot experiments to measure task performance, risk response, decision efficiency, and long-term safety-boundary updates.

We address high-level semantic action selection, not continuous trajectory-level safety guarantees. Risk and uncertainty scores support relative comparisons; they are not calibrated collision probabilities.

# References


[1] I. Singh et al., "ProgPrompt: Generating Situated Robot Task Plans using Large Language Models," *ICRA*, 2023.

[2] Y. Wu et al., "SELP: Generating Safe and Efficient Task Plans for Robot Agents with Large Language Models," *ICRA*, 2025.

[3] N. Shinn et al., "Reflexion: Language Agents with Verbal Reinforcement Learning," *NeurIPS*, 2023.

[4] A. Zhao et al., "ExpeL: LLM Agents Are Experiential Learners," *AAAI*, 2024.

[5] G. Wang et al., "Voyager: An Open-Ended Embodied Agent with Large Language Models," arXiv:2305.16291, 2023.

[6] A. A. Khan et al., "Safety Aware Task Planning via Large Language Models in Robotics," *IROS*, 2025.

[7] Z. Ravichandran et al., "Safety Guardrails for LLM-Enabled Robots," *IEEE RA-L*, 2026.